

CFD Study: The Effect of Radiator Placement on Room Temperature

Chapter 1 – Abstract

This study investigates the influence of radiator placement on the temperature distribution and flow behavior within a room using computational fluid dynamics (CFD) simulations in OpenFOAM. The room is modeled as a simplified rectangular domain, and natural convection is considered as the dominant mechanism for heat transfer.

The simulations are performed under steady-state conditions using the Boussinesq approximation to account for buoyancy effects. Turbulence is modeled with the $k-\epsilon$ model, and heat losses through walls and windows are represented by convective boundary conditions.

Several radiator configurations are analyzed and compared in terms of temperature distribution and flow structure. The results show that the placement of the radiators has only a minor effect on the average room temperature. The maximum temperature difference between the investigated cases is less than 0.2 K.

In contrast, the internal flow structure is strongly influenced by the radiator positioning, leading to different circulation patterns and mixing behaviors. Despite these variations, natural convection ensures a nearly uniform temperature distribution in all configurations.

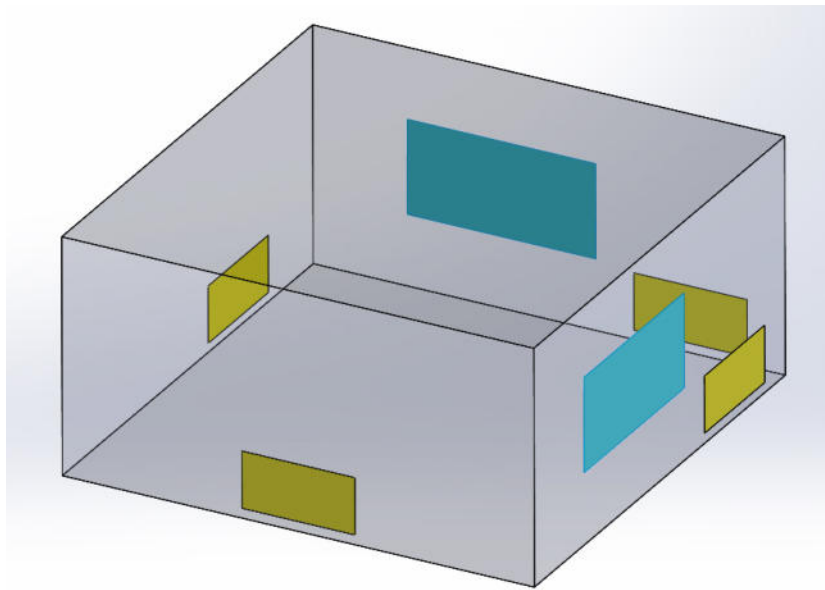
It is concluded that, for the investigated setup, the total heating power is the dominant factor determining the thermal conditions, while radiator placement mainly affects the flow structure rather than the final temperature distribution.

Chapter 2 – Model

The model consists of a rectangular solid with dimensions of 5 m × 5 m × 2.5 m, which is created in SolidWorks.

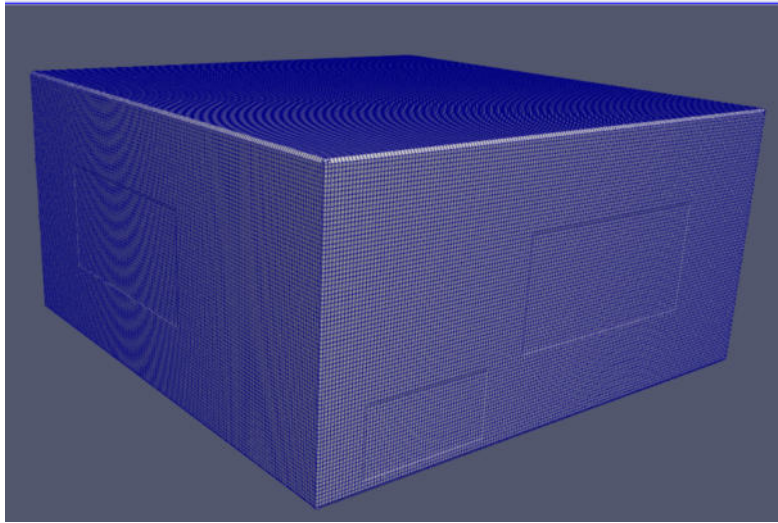
2.1 Generating the geometry

The objective of this study is to use CFD simulation in OpenFOAM to investigate the temperature distribution in a room and to determine how the position of the radiators affects that distribution. The room is created as an assembly. The windows and radiators are imported as flat components. This makes it possible to export all components as STL files using the same coordinate system. To improve the mesh quality in snappyHexMesh, the parts were shifted slightly within the volume (5 mm).



2.2 Meshing

The computational domain is discretized using a structured volume mesh. First, a Cartesian base mesh is generated using blockMesh. Next, snappyHexMesh is used to adapt the mesh to the geometry. However, no targeted local refinement of the mesh is performed, resulting in a largely uniform cell distribution throughout the computational domain. The resulting mesh consists predominantly of hexahedral cells. For the standard configuration, a discretization of $60 \times 60 \times 40$ cells is used, corresponding to a total of approximately 144.000 control volumes. To evaluate the mesh dependency, a mesh study was conducted in which simulations with different mesh resolutions were compared. The results show that the temperature distribution depends only slightly on the mesh, so that the chosen mesh resolution can be considered sufficient for the present investigation.



2.3 Boundary and Initial Conditions

The boundary conditions of the system under consideration are chosen such that the thermally driven flow in the room is realistically represented. The walls of the room are modeled as rigid, impermeable boundaries.

The initial conditions are chosen to be homogeneous. At the start of the simulation, the entire room volume is initialized with a uniform temperature and without an initial flow. The flow motion thus develops exclusively due to the thermal buoyancy forces resulting from the temperature difference between the radiators and the environment.

2.3.1 Velocity

A no-slip condition is applied to the velocity, so that the flow comes to rest at the walls.

2.3.2 Temperature

The wall and the windows are modeled using a convective heat transfer condition in which the heat flux is proportional to the temperature difference between the wall/window and the surrounding environment. The ambient temperature is set to 288 K and the heat transfer coefficients are specified in the following table:

Geometry	Heat transfer coefficient W / (m ² K)	Description
wall	2	well-insulated wall representing low convective heat transfer to the environment
windows	10	window surface representing increased convective heat transfer and higher heat loss

The radiators are modeled as isothermal surfaces with an elevated temperature. They represent the only heat source in the system and drive natural convection in the room. The temperature at the radiators is set to 360 K (87 °C).

2.3.3 Pressure

For the pressure, a reference cell is defined in the computational domain to uniquely solve the system of equations. The pressure gradient results from the flow itself.

2.3.4 Initial Conditions

The initial temperature field in the room is set to 293 K and the initial velocity field is set to zero (0, 0, 0).

2.4 Assumptions and Turbulence

The numerical simulation is performed under steady-state conditions. Transient effects are neglected, as the focus of the investigation lies on the resulting equilibrium state of the temperature and flow field. Buoyancy effects are modeled using the Boussinesq approximation. Density variations are assumed to be small and are only considered in the buoyancy term of the momentum equation. This approach is suitable for the relatively moderate temperature differences present in the investigated scenario.

Turbulence is modeled using the k - ϵ model, which provides a robust and computationally efficient approach for the simulation of indoor airflows. Near-wall effects are treated using wall functions, allowing the use of a relatively coarse mesh without explicitly resolving the boundary layer.

The fluid is assumed to be Newtonian and incompressible, with constant thermophysical properties. Heat transfer occurs through convection and conduction within the fluid, while radiative heat transfer is neglected.

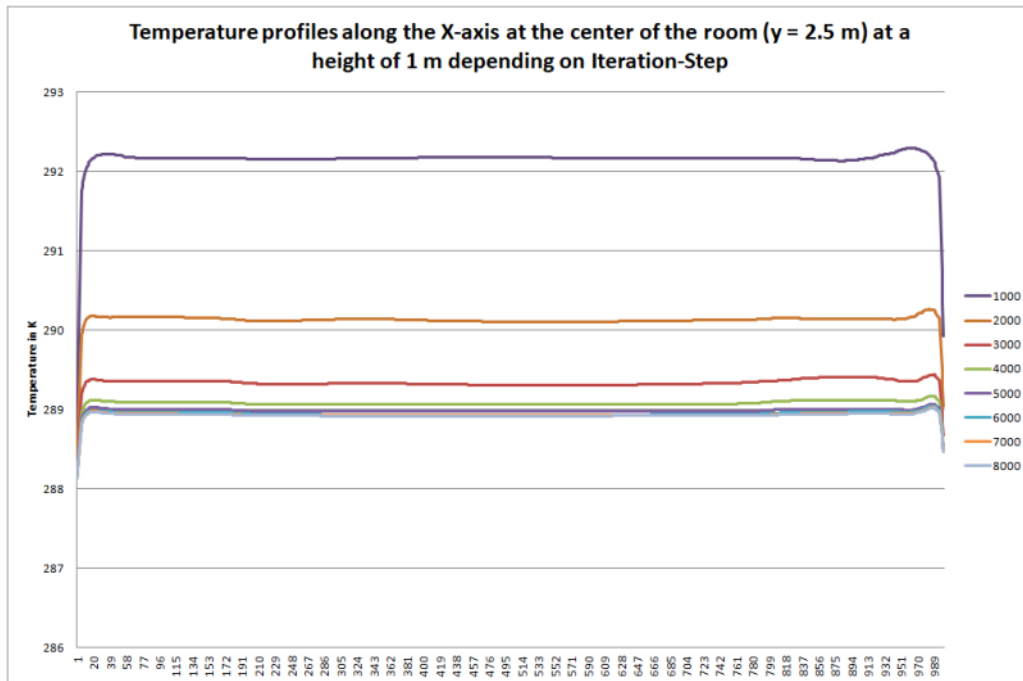
The system is considered closed, with no mass exchange with the surroundings. Air movement is driven solely by natural convection resulting from temperature differences between the heating surfaces and the environment.

Chapter 3 – Numerics

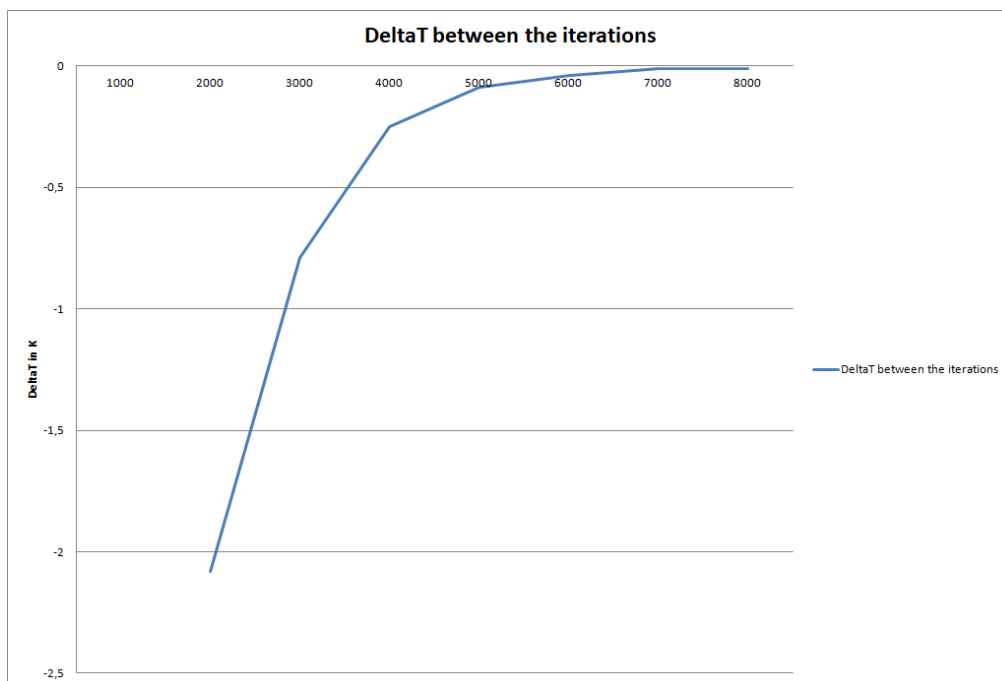
The simulations are performed using the solver `buoyantSimpleFoam`, which is designed for steady-state buoyancy-driven flows. Pressure–velocity coupling is achieved using the SIMPLE algorithm.

Chapter 4 – Convergence and size of Meshing

This study is conducted prior to the actual variant analysis. In this simulation, only one heater was active (position A). The temperature profile is evaluated along a line through the center of the room (x from 0 to 5 m, y = 2.5 m, z = 1 m). The results show that from approximately 6,000 iterations, the temperature changes are negligible. Therefore, an iteration count of 6,000 is considered sufficient.



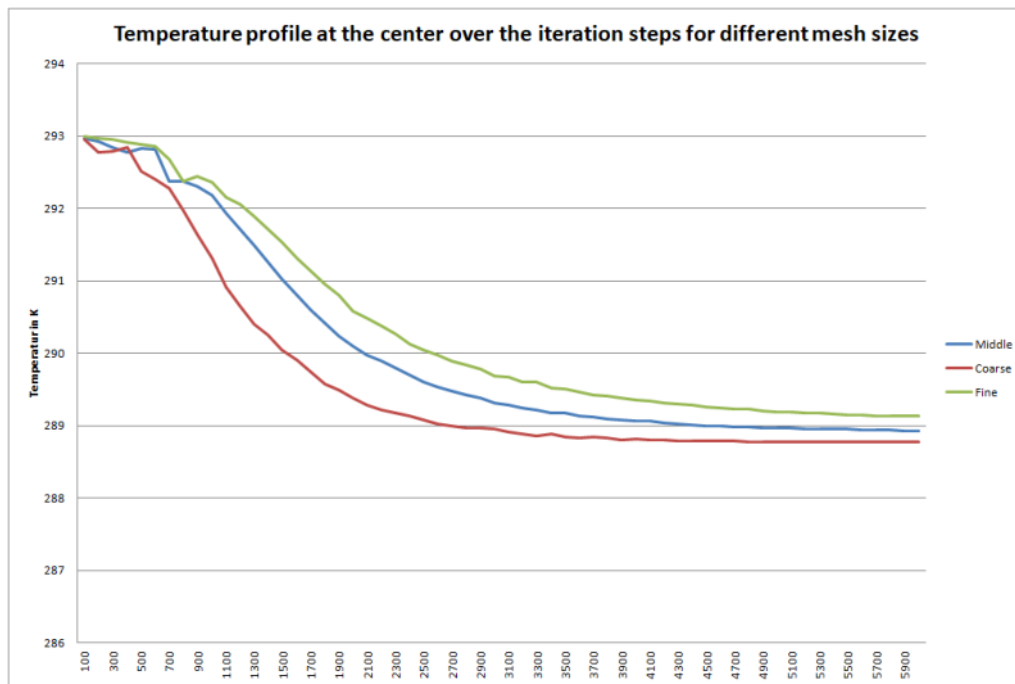
In addition, the following figure shows the temperature difference between the iterations.



It can be seen that, starting at an iteration count of 6.000, there are hardly any noticeable changes (from 6.000 to 7.000 -> ΔT from 0.04 to 0.01 K). Therefore, an iteration count of 6.000 is assumed to be sufficient for the subsequent calculations.

A grid size of 60x60x40 was used as the standard. As part of the grid study, calculations are also being performed using a coarse grid (40x40x30) and a fine grid (80x80x60). This results in the following number of cells:

- Coarse: 40x40x30 = 48.000
- Medium: 60x60x40 = 144.000
- Fine: 80x80x60 = 384.000



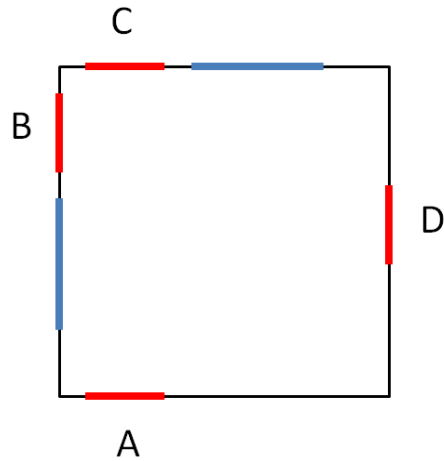
After 6.000 iterations, the following temperatures are obtained at the center:

Mesh size	Temperature [K]
Coarse	288,8
Middle	288,93
Fine	289,1

The medium-resolution grid can therefore be considered sufficiently accurate and is thus used in the investigation scenarios.

Chapter 5 – Variants

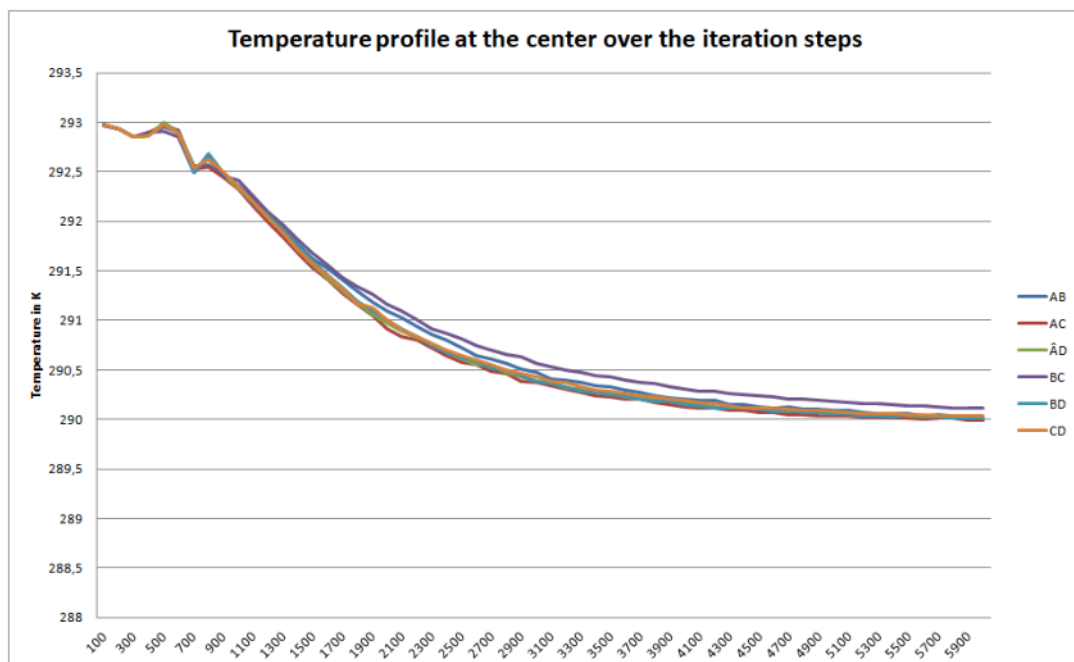
The following illustration shows the possible radiator positions.



With two radiators active at the same time, these four positions result in the following six combinations:

- AB
- AC
- AD
- BC
- BD
- CD

For all of these combinations, 6.000 iteration steps are calculated. The following figure shows the temperature profile at the center point (2.5 m, 2.5 m, 1 m) over the iteration steps for all variants.



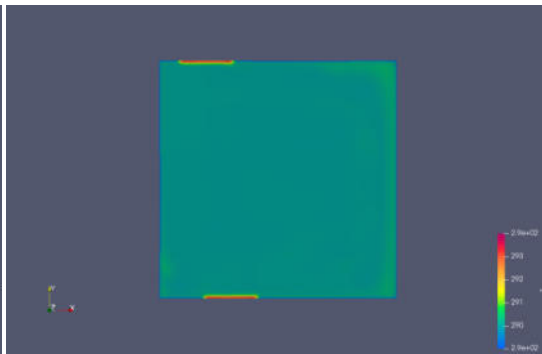
Chapter 6 – Results

An initial comparison of the different radiator configurations is performed using a horizontal slice through the room at a height of $z = 1$ m. The temperature distribution is visualized using a consistent color scale ranging from 288 K to 294 K to ensure comparability between all variants.

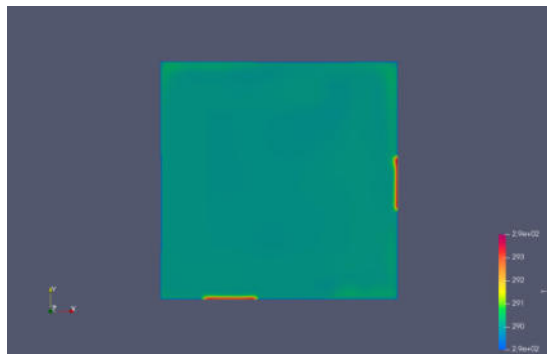
AB



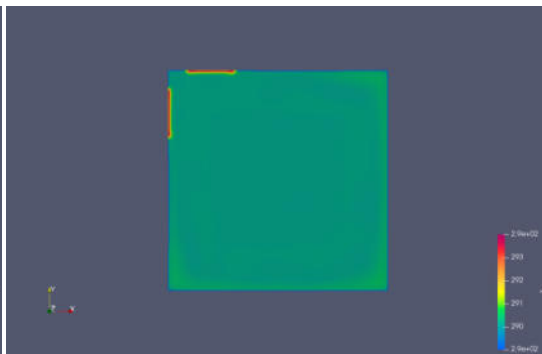
AC



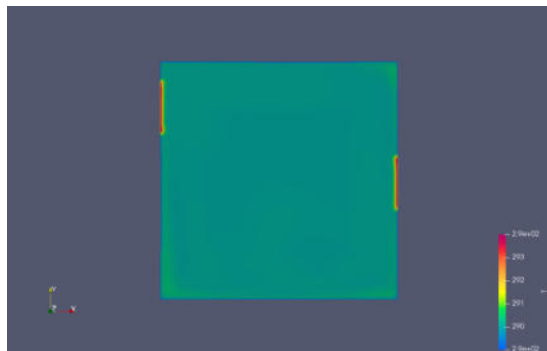
AD



BC



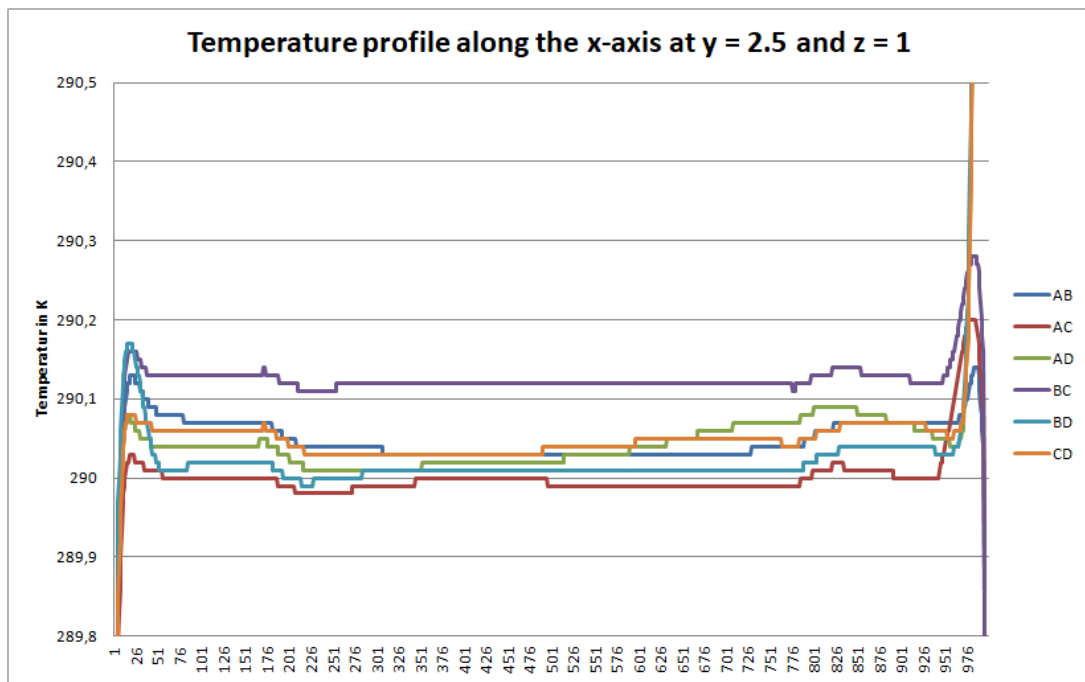
BD



CD



For further analysis, a line plot is evaluated along the X-axis at $y = 2.5$ m and at a height of $z = 1$ m



The results show that the temperature fields of all configurations are very similar. Only minor differences can be observed, particularly in the central region of the room. A quantitative comparison of the temperature at the center point (2.5 m, 2.5 m, 1 m) confirms this observation.

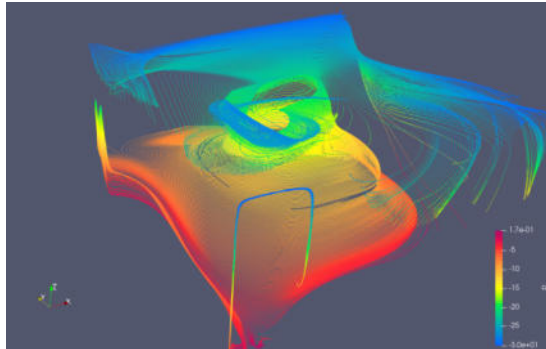
The evaluated temperatures for all configurations in the center point are:

Variant	Temperature in K
AB	290,03
AC	289,99
AD	290,02
BC	290,12
BD	290,01
CD	290,04

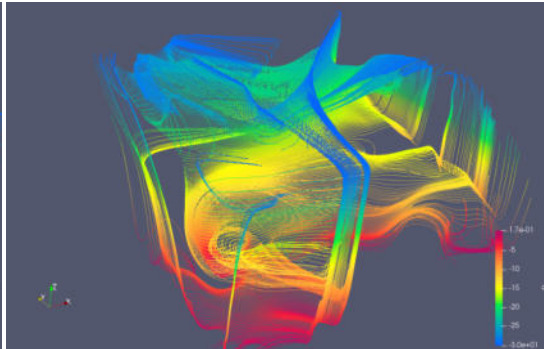
The maximum temperature difference between the variants is approximately 0.13 K, which is negligible compared to the absolute temperature level. Although configuration BC shows a slightly higher temperature at the center, this difference is not considered significant.

In addition to the temperature distribution, the flow field is analyzed using streamline visualizations. These reveal that the radiator placement has a strong influence on the structure of the flow. Depending on the configuration, different recirculation zones and flow patterns are formed within the room.

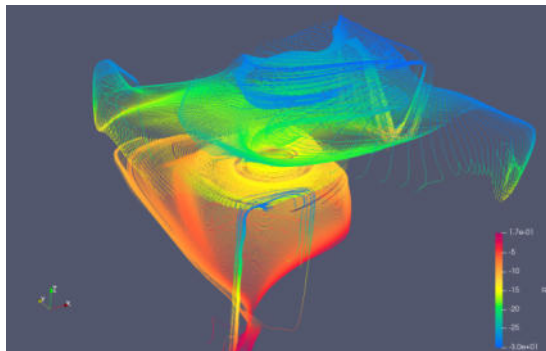
AB



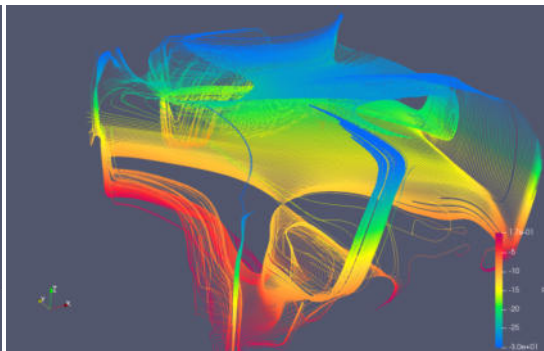
AC



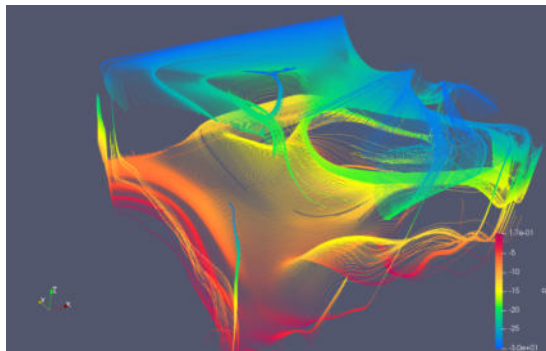
AD



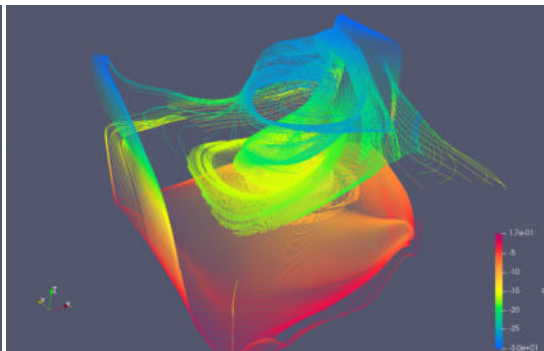
BC



BD



CD



Some configurations exhibit a pronounced large-scale circulation, while others show more localized flow structures. These differences affect the transport paths of the warm air and the mixing behavior within the domain.

However, despite these clearly visible differences in the flow field, the resulting temperature distribution remains largely unchanged. This indicates that natural convection provides sufficient mixing in all configurations, leading to a nearly uniform temperature field throughout the room.

Overall, the results show that the positioning of the radiators has only a minor effect on the average

temperature in the room, while the internal flow structure is significantly influenced by their placement.

These results indicate that, for the investigated setup, the total heating power is the dominant factor for the temperature level, whereas the positioning mainly affects the flow structure rather than the thermal equilibrium.

Future investigations could include transient simulations or more detailed wall modeling to further analyze local thermal effects.